

"ContextOS" — A Universal Personal Memory & Context

Layer

The Core Idea

An open source, **locally-run personal AI memory system** that sits as a middleware layer between *you* and *any* AI model or app. It captures, structures, and selectively injects your personal context (work, habits, preferences, relationships, projects) into any AI interaction — without sending your data to any cloud.

Why It Doesn't Exist Yet (Properly)

- **Mem.ai** is closed source and cloud-dependent
- **Obsidian + AI plugins** are fragmented and require heavy manual curation
- **OpenAI Memory** is siloed to ChatGPT only
- No project owns the *cross-model, privacy-first, programmable* personal context layer

What Makes It Succeed

Technical hook: A local vector + graph hybrid store with a universal context API that any app or model can call via a simple SDK or MCP server. Works with Claude, GPT, Gemini, local LLMs — anything.

Privacy moat: 100% local-first by default. This is the single biggest unmet demand in AI right now. People want AI that knows them without surveillance.

Contributor magnet: Modular "context collectors" — plugins for email, calendar, GitHub, browser history, notes, health data. Every developer has a domain they'd love to contribute.

User magnet: The value compounds daily. The longer you use it, the smarter every AI tool you touch becomes. That's a powerful retention loop.

Architecture Sketch

[Data Sources] → [Context Collectors] → [ContextOS Core]



[Universal Context API / MCP Server]



[Claude / GPT / local LLM / any AI app]

Core stack: **Rust or Go** for performance, **DuckDB + Qdrant** for local hybrid search, **MCP protocol** for AI model integration, **Python SDK** for contributor accessibility.

Why Now

- MCP (Model Context Protocol) just became a real standard — the infrastructure moment has arrived
- Local hardware (Apple Silicon, consumer GPUs) is now powerful enough for always-on local AI middleware
- Growing regulatory pressure (GDPR, EU AI Act) is making cloud-dependent personal AI politically fragile
- No dominant open source project owns this space yet

Comparable Successors to Study

- **Ollama** — succeeded by making local AI *dead simple* → ContextOS should do the same for personal context
- **LangChain** — succeeded by being the glue layer → ContextOS is the *personal data* glue layer
- **Home Assistant** — succeeded with local-first, plugin ecosystem → same playbook, different domain

This sits at the intersection of **privacy**, **developer tooling**, and **daily AI utility** — three of the strongest forces driving open source contribution right now. The team that ships a clean v0.1 with MCP support and 3–4 context collectors (browser, calendar, notes) will own the category.